

Lab Assignment: Constraint Satisfaction Problem

Problem Statement

In this assignment, you are required to implement the 8-Queens Problem as a Constraint Satisfaction Problem (CSP) using Backtracking, MRV, LCV and AC-3. The goal is to compare the efficiency, in terms of the number of steps explored, between plain backtracking, backtracking with heuristics (MRV + LCV), and backtracking enhanced with AC-3.

CSP Representation

- **Variables:** Q1, Q2, Q3, Q4, Q5, Q6, Q7, Q8, where Q_i represents the row selected for the queen in column i .
- **Values:** {1, 2, 3, 4, 5, 6, 7, 8}
- **Constraints:** No two queens can be in the same row or on the same diagonal.
- **Diagonal constraint:** For queens in columns i and j : $|Q_i - Q_j| \neq |i - j|$.

Algorithms to be Used

- Plain Backtracking
- Backtracking with MRV + LCV (Heuristics)
- Backtracking with MRV + LCV + AC-3

Detailed Explanation

1. Start with the plain backtracking implementation. Run it on the 8-Queens problem and record the number of steps taken to reach a solution.
2. Extend the backtracking solver with the MRV (Minimum Remaining Values) and LCV (Least Constraining Value) heuristics. During execution, print which variable was selected using MRV and the order of values chosen using LCV at each decision step.
3. Implement the AC-3 (Arc Consistency Algorithm) for early failure detection. Combine AC-3 with backtracking and the MRV + LCV heuristics. Run the solver and record the number of steps.
4. Compare the step counts for all three approaches. Observe how heuristic search and AC-3 affect the number of explored states.

Required Comparison

Method	Solution Found	Search Steps
Plain Backtracking		
Backtracking + MRV + LCV		
Backtracking + MRV + LCV + AC-3		

Output Requirements

1. Print the final 8-Queens solution as a list of row positions, for example [1, 5, 8, 6, 3, 7, 2, 4].
2. Print a simple chessboard visualization using Q for a queen and . for an empty square.
3. For the heuristic version, print MRV variable selections and the LCV value order at each decision step.

4. For the AC-3 version, show the domain reductions and/or useful information about the arcs processed.
5. Print the comparison table showing the number of steps for all three methods.